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16 August 1961
SUPERSEDING
MIL-G-21032A(SHIPS)
3 March 1959

# MILITARY SPECIFICATION

# GASKETS, METALLIC-ASBESTOS, SPIRAL WOUND (FOR ASA COMMERCIAL FLANGED JOINTS IN PIPING SYSTEMS)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

## 1. SCOPE

1.1 Scope. - This specification covers spiral wound metallic asbestos gaskets for special applications and for pipe line gaskets complete with outer metal retaining rings, to be used with ASA commercial flanges up to 2500 p. s. i. series rating and for temperature of 1050°F.

1.2 Classification. - Gaskets shall be of the following types and series as specified (see 6. 1):

Type I - Gaskets for special applications such as valve bonnets, pumps and other equipment applications (other than boiler manholes and handholes).
Type II - Gaskets for American standard (B16. 5) pipeline flanged joints; with outer metal ring:
- Series 150.
- Series 300.
- Series 400.
- Series 600.
- Series 900.
- Series 1500.
- Series 2500.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
- PPP-B-566 - Boxes, Folding, Paperboard.
- PPP-B-585 - Boxes, Wood, Wirebound.
- PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636 - Boxes, Fiberboard.
- PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
- PPP-B-676 - Boxes, Set-Up, Paperboard.
- PPP-C-843 - Cushioning Material, Cellulosic.

MILITARY
- MIL-B-10377 - Box, Wood, Cleated, Veneer, Paper Overlaid.
- MIL-L-10547 - Liners, Case, Waterproof.
- MIL-P-17667 - Paper; Wrapping, Chemically Neutral.

STANDARDS

MILITARY
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
- MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

FSC 5330

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AMERICAN SOCIETY FOR TESTING MATERIALS

A 167 - Specification for Corrosion-Resisting Chromium Nickel Steel Plate, Sheet and Strip.
A 276 - Specification for Hot-Rolled and Cold-Finished Corrosion-Resisting Steel Bars.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

AMERICAN STANDARDS ASSOCIATION

B16.5 - Steel Pipe Flanges and Flanges and Flanged Fittings.

(Application for copies should be addressed to the American Standards Association, Incorporated, 70 East Forty-fifth Street, New York 17, New York.)

## 3. REQUIREMENTS

3.1 Qualification. - Type II gaskets furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Design. -

3.2.1 Type I gaskets. - Type I gaskets for special applications such as valve bonnets, pumps and other equipment applications (other than boiler manholes and handholes) shall conform to the dimensions required for the application intended and specified (see 6.1), and shall be either 0.125 inch ± 0.005 inch or 0.175 inch ± 0.005 inch initial thickness, as specified (see 6.1). The design shall be such that when the gasket is compressed by a load equivalent to a bolt stress of 30,000 p.s.i., based on cross sectional area at the root of the thread, the thickness shall be as follows:

| Initial thickness Inch | Compressed thickness under 30,000 p.s.i. bolt stress Inch |
|---|---|
| 0.125 ± 0.005 | 0.100 ± 0.005 |
| .175 ± .005 | .130 ± .005 |

3.2.2 Type II, pipe line flange gaskets with outer metal rings. - Type II, gaskets shall consist of a refill snapped in a solid metal outer ring with a sliding fit. The ring shall have a thickness of 1/8 ± 0.006 inch. The ring shall have a V-shaped groove or other satisfactory retaining feature machined on the inside periphery to fit the outside periphery of the refill. The design shall be such that the gasket can be compressed without damage to 0.119 inch thickness. Gaskets shall conform to tables I through IV.

3.2.2.1 Refills for all series of type II gaskets. - Refills shall be of the inside and outside dimensions specified in tables I through IV, inclusive.

3.3 Materials. -

3.3.1 Metal strip. - The metal strip used in the gasket refills shall be of the following materials:

(a) Corrosion resisting steel, AISI type 347, cold rolled and annealed, in accordance with ASTM Specification A167.
(b) Corrosion resisting steel, AISI type 316L, in accordance with ASTM Specification A276.
(c) Corrosion resisting steel, AISI type 309S, in accordance with the applicable requirements of ASTM Specification A276, except that type 309S shall be stabilized with columbium (and tantalum not exceeding 0.10 percent) and their combination to be not less than 10 times the carbon content.

3.3.1.1 Special application. - For special applications other materials shall be specified or specifically approved by the bureau or activity concerned.

3.3.2 Metal outer ring. - The metal outer ring shall be of soft carbon steel. At the option of the manufacturer a protective coating may be applied to the metal outer ring. The resulting dimensions shall conform to tables I through IV.

3.4 Construction. - The gaskets shall be composed of a single strip of metal wound in spiral dovetail shape, starting on the inside and working outward ply by ply to the desired size. Between some of the plies shall be a cushion of asbestos paper (see tables I through IV) either a continuous single length or in two lengths depending upon the design. The asbestos paper may be either a single strip or in two strips. Spot welds on the gasket inside diameter shall be not less than three in number and approximately spaced at 2 inches maximum spacing. The outside weld shall consist of not less than three spot welds.

3.4.1 Service. - Complete gaskets and refills specified in tables I through IV are suitable for temperatures up to 1050°F, and unless otherwise specified, may be used for pressure and temperature combinations in accordance with ASA Publication B16.5.

3.4.2 Metal strip. - The metal strip shall be not less than 0.007 and not more than 0.009 inch in thickness.



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Table I - Gasket characteristics for series 150 p.s.i. ASA flanges.

| I.p.s. | Pipe o.d. | Gasket refills (see figure - 1) | | Number of plies | | | Metal outer ring o.d. | Test 3/ load |
|---|---|---|---|---|---|---|---|---|
| | | | | Metal 2/ | | | | |
| | | I.d. | O.d. | Inside periphery ± 1/2 turn | Outside periphery ± 1/2 turn | Total (MIN) | | |
| Inches | Inches | Inches 1/ | Inches 1/ | | | | Inches | Pounds |
| 1/4 | 0.540 | 1/2 | 27/32 | - | - | -- | 1-3/4 | 12,600 |
| 1/2 | .840 | 3/4 | 1-7/32 | 6 | 3 | 12 | 1-7/8 | 12,600 |
| 3/4 | 1.050 | 1 | 1-17/32 | 6 | 3 | 14 | 2-1/4 | 12,600 |
| 1 | 1.315 | 1- 1/4 | 1-27/32 | 3 | 3 | 9 | 2-5/8 | 12,600 |
| 1- 1/4 | 1.660 | 1-11/16 | 2-13/32 | 3 | 3 | 10 | 3 | 15,100 |
| 1- 1/2 | 1.900 | 2 | 2-23/32 | 3 | 3 | 11 | 3-3/8 | 15,100 |
| 2 | 2.375 | 2- 9/16 | 3-13/32 | 3 | 3 | 11 | 4-1/8 | 24,200 |
| 2-1/2 | 2.875 | 3- 1/16 | 3-27/32 | 3 | 3 | 12 | 4-7/8 | 24,200 |
| 3 | 3.500 | 3- 3/4 | 4-23/32 | 3 | 3 | 12 | 5-3/8 | 24,200 |
| 3-1/2 | 4.000 | 4- 1/8 | 5- 7/32 | 3 | 3 | 14 | 6-3/8 | 48,500 |
| 4 | 4.500 | 4- 3/4 | 5-27/32 | 3 | 3 | 14 | 6-7/8 | 48,500 |
| 5 | 5.563 | 5-13/16 | 6-31/32 | 3 | 3 | 15 | 7-3/4 | 72,500 |
| 6 | 6.625 | 6- 7/8 | 8- 7/32 | 3 | 3 | 15 | 8-3/4 | 72,500 |
| 8 | 8.625 | 8- 7/8 | 10-13/32 | 3 | 3 | 17 | 11 | 72,500 |
| 10 | 10.750 | 10-13/16 | 12-15/32 | 3 | 3 | 18 | 13-3/8 | 151,000 |
| 12 | 12.750 | 12- 7/8 | 14-23/32 | 3 | 3 | 21 | 16-1/8 | 151,000 |
| 14 o.d | 14.000 | 14- 1/4 | 15-31/32 | 3 | 3 | 17 | 17-3/4 | 198,000 |
| 16 o.d | 16.000 | 16- 1/4 | 18- 7/32 | 3 | 3 | 21 | 20-1/4 | 264,000 |

1/ Tolerance plus or minus 1/64 inch on the inside diameter and plus or minus 1/32 inch on the outside diameter.
2/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.
3/ Corresponds to bolting loads of 30,000 p.s.i. unit stress, except for gasket sizes 1 inch and smaller, which are based upon unit bolt stress of 25,000 p.s.i. at thread root area. When under this compressive test load, the gasket thickness shall be 0.130 inch plus or minus 0.005 inch.

3.4.3 Gasket thickness. - The thickness of the finished gasket (before compression) shall be 0.175 ± 0.005 inch for type II gaskets. The thickness of the type I gaskets shall be as specified in 3.2.1.

3.4.4 Plies. - A ply shall consist of one 360 degree turn of metal strip. Plies shall be counted adjacent to a terminal weld. An asbestos ply shall consist of one 360 degree turn of asbestos strip(s) depending upon the design.

3.4.5 Gaskets shall be capable of passing the compression load tests indicated in 4.5.1.

3.5 Marking. -

3.5.1 Type I gaskets. - Each type I gasket shall be marked with a sturdy paper tag (of tag stock) securely attached thereto, which shall contain thereon as much of the following information as possible: the equipment manufacturer's name, type of equipment and model number, and gasket part number; gasket manufacturer's name, customer's order number and any other data required to clearly identify the end use of the gasket.

3.5.2 Type II gaskets. - Each type II gasket shall be steel-stamped clearly on one side of the outer ring to show the manufacturer's name, pipe size, the pressure series and "ASA-MIL" to indicate that the gasket is for ASA flanges and also meets the requirements of this specification (example - "2-600 ASA-MIL").

3.5.3 Refills. - When refills are purchased separate from the rings, each refill shall be identified using a piece of pressure sensitive tape, by enclosing each refill in a paper or plastic bag, or by other means approved by the bureau or activity concerned, having the manufacturer's name, pipe size, pressure series and "ASA-MIL" printed thereon. If pressure sensitive tape is applied, the tape shall be looped through the refill and brought back upon itself for firm retention. If a paper or plastic bag is used, it shall be stapled or otherwise closed to insure that the refill is retained in the bag.



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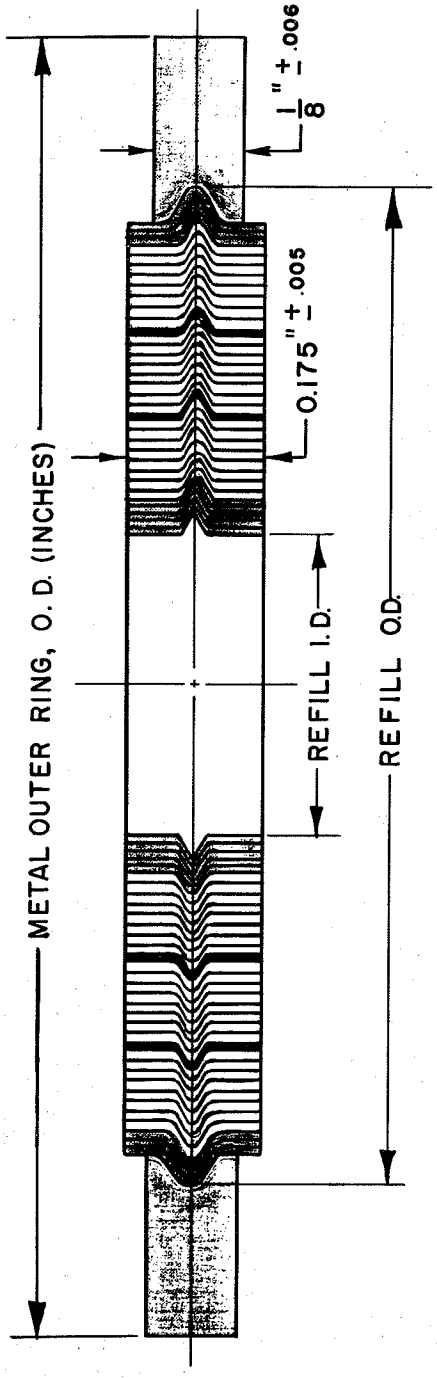



Figure 1 - Gasket.



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Table II - Type II gasket characteristics for series 300, 400 and 600 ASA flanges.

| I. p. s. | Pipe o. d. | Gasket refills (see figure - 1) | | Number of plies | | | Metal outer ring o. d., inches | | | Test load 3/ |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Metal 2/ | | | | | | |
| | | I. d. inch 1 | O. d. inch 1/ | Inside periphery ± 1/2 turn | Outside periphery ± 1/2 turn | Total (Min.) | Series 300 | Series 400 | Series 600 | |
| Inches | Inches | | | | | | | | | Pounds |
| 1/4 | 0.540 | 1/2 | 27/32 | A | 3 | 9 | 1-3/4 | 1-3/4 | 1-3/4 | 12,600 |
| 1/2 | .840 | 3/4 | 1-7/32 | 6 | 3 | 12 | 2-1/8 | 2-1/8 | 2-1/8 | 12,600 |
| 3/4 | 1.050 | 1 | 1-17/32 | 6 | 3 | 14 | 2-5/8 | 2-5/8 | 2-5/8 | 20,200 |
| 1 | 1.315 | 1-1/4 | 1-27/32 | 6 | 3 | 14 | 2-7/8 | 2-7/8 | 2-7/8 | 20,200 |
| 1-1/4 | 1.660 | 1-11/16 | 2-11/32 | 6 | 3 | 14 | 3-1/4 | 3-1/4 | 3-1/4 | 24,200 |
| 1-1/2 | 1.900 | 2 | 2-23/32 | 6 | 3 | 15 | 3-3/4 | 3-3/4 | 3-3/4 | 36,200 |
| 2 | 2.375 | 2-9/16 | 3-11/32 | 6 | 3 | 15 | 4-3/8 | 4-3/8 | 4-3/8 | 48,400 |
| 2-1/2 | 2.875 | 2-15/16 | 3-27/32 | 6 | 3 | 16 | 5-1/8 | 5-1/8 | 5-1/8 | 72,500 |
| 3 | 3.500 | 3-3/4 | 4-23/32 | 6 | 3 | 18 | 5-7/8 | 5-7/8 | 5-7/8 | 72,500 |
| 3-1/2 | 4.000 | 4-1/8 | 5-7/32 | 6 | 3 | 20 | 6-1/2 | 6-3/8 | 6-3/8 | 101,000 |
| 4 | 4.500 | 4-3/4 | 5-27/32 | 6 | 3 | 20 | 7-1/8 | 7 | 7-5/8 | 101,000 |
| 4-1/2 | 5.000 | 5-5/16 | 6-15/32 | 6 | 3 | 22 | 7-3/4 | 7-5/8 | ----- | 132,000 |
| 5 | 5.563 | 5-13/16 | 6-31/32 | 6 | 3 | 22 | 8-1/2 | 8-3/8 | 9-1/2 | 132,000 |
| 6 | 6.625 | 6-7/8 | 8-7/32 | 6 | 3 | 24 | 9-7/8 | 9-3/4 | 10-1/2 | 198,000 |
| 8 | 8.625 | 8-7/8 | 10-11/32 | 6 | 3 | 27 | 12-1/8 | 12 | 12-5/8 | 262,000 |
| 10 | 10.750 | 10-13/16 | 12-15/32 | 6 | 3 | 32 | 14-1/4 | 14-1/8 | 15-3/4 | 446,000 |
| 12 | 12.750 | 12-7/8 | 14-23/32 | 6 | 3 | 32 | 16-5/8 | 16-1/2 | 18 | 557,000 |
| 14 o. d. | 14.000 | 14-1/4 | 15-31/32 | 6 | 3 | 32 | 19-1/8 | 19 | 19-3/8 | 693,000 |
| 16 o. d. | 16.000 | 16-1/4 | 18-7/32 | 6 | 3 | 35 | 21-1/4 | 21-1/8 | 22-1/4 | 840,000 |
| 18 o. d. | 18.000 | 18-1/2 | 20-23/32 | -- | -- | -- | 23-1/2 | 23-3/8 | 24-1/8 | 1,008,000 |

1/ Tolerance plus or minus 1/64 inch on the inside diameter and plus or minus 1/32 inch on the outside diameter.
2/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.
3/ Corresponds to bolting load of 30,000 p. s. i. unit stress, except for gasket sizes 1 inch and smaller, which are based upon unit bolt stress of 25,000 p. s. i. at the thread root area. When under this compressive test load, the gasket thickness shall be 0.130 inch plus or minus 0.005 inch.





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Table III - Type II gasket characteristics for series 900 and 1500 p. s. i. ASA flanges.

| I. p. s. | Pipe o. d. | Gasket refills series 900 and 1500 ASA flanges (see fig. 1) | | Number of plies | | | Metal outer ring o. d. inch | | Test load 3/ pounds | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Metal 2/ | | | | | Series 900 | Series 1500 |
| | | I. d. | O. d. | Inside periphery ± 1/2 turn | Outside periphery ± 1/2 turn | Total (Min) | 900 | 1500 | | |
| Inches | Inches | Inches 1/ | Inches 1/ | | | | P. s. i. | P. s. i | | |
| 1/4 | 0.540 | ----- | ------ | -- | -- | -- | ----- | ----- | ------ | ------ |
| 1/2 | .840 | 3/4 | 1-7/32 | 6 | 3 | 15 | 2-1/2 | 2-1/2 | 30, 200 | 30, 200 |
| 3/4 | 1.050 | 1 | 1-17/32 | 6 | 3 | 16 | 2-3/4 | 2-3/4 | 30, 200 | 30, 200 |
| 1 | 1.315 | 1-1/4 | 1-27/32 | 6 | 3 | 18 | 3-1/8 | 3-1/8 | 41, 900 | 41, 900 |
| 1-1/4 | 1.660 | 1-11/16 | 2-11/32 | 6 | 3 | 19 | 3-1/2 | 3-1/2 | 50, 300 | 50, 300 |
| 1-1/2 | 1.900 | 2 | 2-23/32 | 6 | 3 | 21 | 3-7/8 | 3-7/8 | 66, 100 | 66, 100 |
| 2 | 2.375 | 2-9/16 | 3-11/32 | 6 | 3 | 22 | 5-5/8 | 5-5/8 | 101, 000 | 101, 000 |
| 2-1/2 | 2.875 | 2-15/16 | 3-27/32 | 6 | 3 | 25 | 6-1/2 | 6-1/2 | 132, 000 | 132, 000 |
| 3 | 3.500 | 3-3/4 | 4-23/32 | 6 | 3 | 27 | 6-5/8 | 6-7/8 | 101, 000 | 175, 000 |
| 4 | 4.500 | 4-3/4 | 5-27/32 | 6 | 3 | 29 | 8-1/8 | 8-1/4 | 175, 000 | 223, 000 |
| 5 | 5.563 | 5-13/16 | 6-31/32 | 6 | 3 | 31 | 9-3/4 | 10 | 223, 000 | 337, 000 |
| 6 | 6.625 | 6-7/8 | 8-7/32 | 6 | 3 | 35 | 11-3/8 | 11-1/8 | 262, 000 | 416, 000 |
| 8 | 8.625 | 8-7/8 | 10-11/32 | 6 | 3 | 38 | 14-1/8 | 13-7/8 | 416, 000 | 605, 000 |
| 10 | 10.725 | 10-13/16 | 12-15/32 | 6 | 3 | 40 | 17-1/8 | 17-1/8 | 554, 000 | 829, 000 |
| 12 | 12.750 | 12-7/8 | 14-23/32 | 6 | 3 | 42 | 19-5/8 | 20-1/2 | 693, 000 | 1, 273, 000 |

1/ Tolerance plus or minus 1/64 inch on the inside diameter and plus or minus 1/32 inch on the outside diameter.
2/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.
3/ Corresponds to bolting load of 30, 000 p. s. i. unit stress, except for gasket sizes 1 inch and smaller, which are based upon unit bolt stress of 25, 000 p. s. i. at the thread root area. When under this compressive test load, the gasket thickness shall be 0. 130 inch plus or minus 0. 005 inch.

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Table IV - Type II gasket characteristics for series 2500 p. s. i. ASA flanges.

| I. p. s. | Pipe o. d. | Gasket refills (see figure - 1) | | Number of plies | | | Metal outer ring o. d. | Test load 3/ |
|---|---|---|---|---|---|---|---|---|
| | | | | Metal 2/ | | | | |
| | | I. d. | O. d. | Inside periphery ± 1/2 turn | Outside periphery ± 1/2 turn | Total (Min.) | | |
| Inches | Inches | Inches 1/ | Inches 1/ | | | | Inches | Pounds |
| 1/2 | 0.840 | 3/4 | 1-7/32 | 6 | 3 | 15 | 2-3/4 | 30,200 |
| 3/4 | 1.050 | 1 | 1-17/32 | 6 | 3 | 16 | 3 | 30,200 |
| 1 | 1.315 | 1-1/4 | 1-27/32 | 6 | 3 | 18 | 3-3/8 | 41,900 |
| 1-1/4 | 1.660 | 1-9/16 | 2-11/32 | 6 | 3 | 19 | 4-1/8 | 55,100 |
| 1-1/2 | 1.900 | 1-7/8 | 2-23/32 | 6 | 3 | 24 | 4-5/8 | 72,800 |
| 2 | 2.375 | 2-5/16 | 3-11/32 | 6 | 3 | 29 | 5-3/4 | 110,000 |
| 2-1/2 | 2.875 | 2-3/4 | 3-27/32 | 6 | 3 | 32 | 6-5/8 | 146,000 |
| 3 | 3.500 | 3-5/8 | 4-23/32 | 6 | 3 | 32 | 7-3/4 | 188,000 |
| 4 | 4.500 | 4-5/8 | 5-27/32 | 6 | 3 | 35 | 9-1/4 | 281,000 |
| 5 | 5.563 | 5-7/8 | 6-31/32 | 6 | 3 | 36 | 11 | 396,000 |
| 6 | 6.625 | 6-3/4 | 8-7/32 | 6 | 3 | 38 | 12-1/2 | 636,000 |
| 8 | 8.625 | 8-1/2 | 10-11/32 | 6 | 3 | 41 | 15-1/4 | 955,000 |
| 10 | 10.750 | 10-5/8 | 12-15/32 | 6 | 3 | 41 | 18-3/4 | 1,545,000 |
| 12 | 12.750 | 12-3/4 | 14-23/32 | 6 | 3 | 50 | 21-5/8 | 1,893,000 |

1/ Tolerance plus or minus 1/64 inch on the inside diameter and plus or minus 1/32 inch on the outside diameter.
2/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.
3/ Corresponds to bolting load on 25,000 p. s. i. unit stress. When under this compressive test load, the gasket thickness shall be 0.130 inch plus or minus 0.005 inch.

3.6 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. 1/ - Qualification examination and testing of type II gaskets shall be conducted at a laboratory satisfactory to the Bureau of Ships. A supplier may have his gaskets qualified in any pressure series by passing the specified tests for the series concerned. Qualification examination and tests shall consist of those specified in 4.4 and 4.5.1.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

4.2.1 Samples for qualification tests. - For purposes of qualification testing, three gaskets of each of the following series and sizes shall be furnished:

| Pressure series | Sample sizes required |
|---|---|
| 150 | 2 inch, 8 inch, 12 inch |
| 1 300, 400 and 600 | 3 inch, 6 inch, 10 inch |
| 900 | 2/ 6 inch, 10 inch |
| 1500 | 2 inch, 4 inch, 6 inch |
| 2500 | 2 inch, 3 inch, 5 inch |

1/ Since the gasket body (or refill) is identical for series 300, 400 and 600, only one set of samples for these series is required.
2/ Qualification of the series 1500, 2 inch size sample will also constitute conformance of the series 900, 2 inch size.



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4.3 Sampling for lot acceptance inspection. -

4.3.1 Lot. - All gaskets of the same type and size offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.3.2 Sampling for visual and dimensional examination. - Sample gaskets shall be selected at random from each lot in accordance with table V for the examination specified in 4.4.

4.3.3 Sampling for compression test. - Sample gaskets shall be selected from each lot in accordance with table VI for the tests specified in 4.5.1. If any gasket fails the compression test, the entire lot shall be rejected.

Table V - Sampling for visual and dimensional examination.

| Lot size (number of gaskets) | Sample size (number to be examined) | Acceptance number (Defectives) | Rejection number (Defectives) |
|---|---|---|---|
| 2 to 10 | All | - | - |
| 11 to 25 | 10 | 0 | 1 |
| 26 to 40 | 15 | 0 | 1 |
| 41 to 110 | 25 | 1 | 2 |
| 111 to 300 | 35 | 1 | 2 |
| 301 to 500 | 50 | 2 | 3 |
| 501 to 800 | 75 | 3 | 4 |
| 801 to 1300 | 110 | 4 | 5 |
| 1301 and over | 150 | 5 | 6 |

Table VI - Sampling for compression test.

| Lot size (Number of gaskets) | Sample size (Number to be tested) |
|---|---|
| 1 to 25 | 1 |
| 26 to 180 | 2 |
| 181 to 500 | 3 |
| 501 to 800 | 5 |
| 801 to 1,300 | 7 |
| 1,301 to 3,200 | 10 |
| 3,201 to 8,000 | 15 |
| 8,001 to 22,000 | 25 |
| 22,001 and over | 35 |

4.4 Visual and dimensional examination. - Each of the sample gaskets selected in accordance with 4.3.2 shall be examined and measured to verify conformance with all of the requirements of this specification which do not include the compression tests. Examination shall be conducted as specified in table VII. The number of plies shall be counted, a magnifying glass being used to insure accuracy of the count.

4.5 Test procedures. -

4.5.1 Compression load test. -

4.5.1.1 The test shall consist of subjecting the gaskets to compression in a Baldwin Southwark, or equal, hydraulic compression machine of suitable capacity and measuring the thickness of the gaskets while under the tests loads specified in 3.2.1 for type I gaskets and tables I through IV, as applicable, for type II gaskets.

4.5.1.2 The gaskets shall be tested between steel test plates, the surfaces of which shall have a smooth finish machined with a circular lay (concentric or spiral) having a minimum of 32 serrations per inch of face width with depth of serrations not exceeding 0.003 inch of an RHR finish of from 250 to 500. For special installations involving radioactive service or hazardous fluids where a finer finish is required, and so indicated in the contract or order, a maximum RHR of 125 shall be used. The test plates shall be centrally located under the ball and socket head of the test machine. The ball and socket head shall be used to equally distribute the load applied to the gasket. By means of inverted T straps, the upper compression plate shall be suspended and held in place when inserting a gasket. Four dial indicators graduated in 1/1000 inch divisions, located 90 degrees apart and mounted on the upper compression plate are employed for measuring the thickness of the gasket.

4.5.1.3 During the test, the increment of load shall be applied until a load corresponding to the test load in accordance with 3.2.1 for type I gaskets and 3.2.2 for type II gaskets is obtained for the gasket under test. The maximum rate of loading allowed will be 2,000 pounds per second.

4.5.1.4 The steel test plates shall be placed in contact with each other and under an applied load of 3,000 pounds, the dial indicators shall be set so that a zero reading is obtained with the maximum travel of the indicator plunger. The plates shall then be separated, a gasket centrally located between plates and the upper plate brought



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into contact with the gasket but no load applied. The dial indicators shall then be read, and the average of the four readings should closely check the original micrometer thickness of the gasket. The load shall then be applied and the rate of loading depends on the size of gasket and the maximum load to be applied. At each interval, the specified load shall be maintained only for a sufficient time, 5 to 10 seconds, to obtain the dial indicator readings. By means of reading the dial indicator scales counter clockwise instead of clockwise the actual thickness of the gasket shall be directly obtained. The average of the four dial indicator readings shall be considered to be the thickness of the gasket. After completion of the test, the load shall be released but the upper compression plate shall be left in contact with the gasket for approximately 10 seconds and the dial indicator readings shall be taken to determine the recovery of the gaskets. The test plates shall then be separated, the gasket removed from the test machine, measured for thickness by micrometers, examined for buckling on the inner periphery, condition of welds, and binding of the gasket filler in the outer metal ring. It may be noted that the final micrometer thickness and the released average dial indicator thickness of the gasket will be approximately the same providing the dial indicators are in proper adjustment. Occasionally between test runs the zero setting of the dial indicators shall be checked with the test plates in contact with one another. No buckling on the inner edge of the gasket shall be allowed. Gaskets shall at least recover 0.010 inch over the compressed value obtained in the compression test specified in 4.5.1.3.

Table VII - Classification of defects in accordance with Standard MIL-STD-105.

| Categories | Defects |
|---|---|
| Critical | None defined. |
| Major | |
| 101 | Type of gasket not specified. |
| 102 | Number of metal plies less than specified minimum. |
| 103 | Number of asbestos plies less than the specified minimum or plies less than a 360 degree turn each. |
| 104 | Spot welds (on inside and outside diameters) missing, not the required number, incorrectly spaced, or spot welds defective. |
| 105 | Thickness of metal strip not within the specified maximum and minimum values, or metal strip not continuous (one piece). |
| 106 | Thickness of finished gasket not within specified tolerances. |
| 107 | Gasket refill not a sliding fit in the metal ring, or metal ring not provided with a satisfactory means of retaining the refill. |
| 108 | Inside or outside diameter of refill not within the specified tolerances. |
| 109 | Failures of gasket to conform to the dimensions specified for valve bonnets, pumps and other applications (type I only). |
| 110 | Identification markings missing, illegible, incorrect or not as specified. |

4.5.2 Possible test failures. - Possible test failures are defined as follows:

(a) Gasket buckled excessively on inner periphery to the extent that metal plies are separated.
(b) Welding failed.
(c) Gasket thickness recovered less than 0.010 inch after release of compression test load.

4.5.3 Chemical analysis - Mill certification of materials may be accepted by the Government inspector in lieu of performing chemical and physical tests. Such certification should contain the results of the chemical check analysis of the heat from the steel obtained, the heat number and date to indicate that the stock is not over 1 year old. When considered necessary by the inspector, the samples selected in accordance with 4.2.1 and 4.3 shall be subjected to a chemical analysis to determine conformance with 3.3.1.

4.6 Comparison examination and tests (for type II gaskets only). - Comparison examination and tests shall be conducted to verify continuing quality of the approved product and compliance with the specification requirements. These tests will be on samples representing those series and sizes which were tested for qualification and approved and will be conducted on samples supplied by the manufacturer, at no cost to the Government. Comparison examination and tests are not required more often than every two years unless the inspector has reason to consider it necessary. Comparison examination and tests shall be those specified



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in 4.4 and 4.5.1 plus any others specifically requested by the Government inspector.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- Gaskets and refills, quantity as specified (see 6.1), shall be packaged in boxes conforming to Specification PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676, at the option of the contractor. When specified (see 6.1) spacers shall be provided on type II gaskets between outer metal rings. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.1 Bundling.- When specified (see 6.1), gaskets shall be bundled in the quantities, tied and wrapped in Kraft paper as follows:

5.1.1.1.1 Type I gaskets (without outer metal rings).-

| Size (O.D.) | Quantity and tying |
|---|---|
| Up to and including 6-inches | 25 pieces. |
| Over 6 inches to 12 inches, inclusive. | 10 pieces, tied in 2 equidistant places. |
| Over 12-inches to 15-inches, inclusive. | 10 pieces, tied in 3 equidistant places. |
| Over 15 inches. | 5 pieces, tied in 4 equidistant places. |

5.1.1.1.2 Type II gaskets (with outer metal rings).-

| Size (O.D.) | Quantity and tying |
|---|---|
| Up to and including 5 inches. | 25 pieces. |
| Over 5 inches to 8 inches, inclusive. | 25 pieces, tied in 2 equidistant places. |
| Over 8 inches to 11 inches, inclusive. | 25 pieces, tied at 4 equidistant places. |
| Over 11 inches to 16 inches, inclusive. | 10 pieces, tied at 4 equidistant places. |
| Over 16 inches. | 5 pieces, tied at 4 equidistant places. |

Spacers shall be provided between outer metal rings. Wrapping paper shall conform to Specification MIL-P-17667.

5.1.1.2 Bulk.- When specified (see 6.1), gaskets shall be bulk packed directly into shipping containers as specified in 5.2.1 and 5.2.2.

5.1.2 Level C.- Gaskets and refills shall be packaged in a manner sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2 Packing.-

5.2.1 Level A.- Gaskets and refills, packaged as specified (see 6.1), shall be packed in fiberboard boxes conforming to class 2, grade 3 of Specification PPP-B-636. Alternatively, gasket and refills may be packed in containers conforming to any one of the following specifications at the option of the contractor:

| Specification | Type or class |
|---|---|
| PPP-B-585 | Class 3 use |
| PPP-B-591 | Overseas type |
| PPP-B-601 | Overseas type |
| PPP-B-621 | Class 2 |
| MIL-B-10377 | Overseas type |

Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for class 2 fiberboard boxes conforming to Specification PPP-B-636 may be omitted provided all center and edge seams and manufacturer's joints are waterproofed in accordance with the appendix to Specification PPP-B-636. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2 Level B- Gaskets and refills, packaged as specified (see 6.1), shall be packed in fiberboard boxes having a minimum bursting strength of 275 p.s.i., conforming to Specification PPP-B-636, class 1. Alternatively, gaskets and refills may be packed in containers conforming to any one of the following Specifications at the option of the contractor:

| Specification | Type of class |
|---|---|
| PPP-B-585 | Class 1 or 2 use |
| PPP-B-591 | Domestic type |
| PPP-B-601 | Domestic type |
| PPP-B-621 | Class 1 |
| MIL-B-10377 | Domestic type |

Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.3 Level C.- Gaskets of like types, series and size, packaged as specified (see 6.1), shall be packed in containers which will insure acceptance by



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common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2.4 Filling of voids.- Voids or cavities in the shipping containers, caused by wrapped bundles or bulk packing, shall be filled with cushioning materials to prevent shifting of contents and damage during handling, shipment and storage. The use of loose excelsior, shredded newspaper and similar materials which have not been refined for noncorrosiveness shall not be used unless these materials are enclosed in a sealed waterproof barrier material. Cushioning materials shall conform to Specification PPP-C-843 or other materials that are noncorrosive to the packed item.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, series and size of gaskets required.
(c) For type II gaskets specify whether refills or complete gaskets are required. If refills are required, specify type, series and size required.
(d) Quantity per unit package (see 5.1.1).
(e) Selection of applicable level of packaging and packing (see 5.1 and 5.2).
(f) For type I gaskets, specify the type and model of equipment on which used, the equipment manufacturer, the initial equipment manufacturer's part number for the gasket, the gasket thickness (see 3.2.1) and any other pertinent information.

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 21032 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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